Schizophrenia Research 272 (2024) 1–11

Contents lists available at ScienceDirect

Schizophrenia Research

journal homepage: www.elsevier.com/locate/schres

Efficacy and tolerability of antipsychotic polypharmacy for schizophrenia spectrum disorders. A systematic review and meta-analysis of individual patient data

Marc W.H. Lochmann van Bennekom [a,z,*], Joanna IntHout [b], Harm J. Gijsman [c], Berna B.K. Akdede [d], A. Elif Anıl Yağcıoğlu [e], Thomas R.E. Barnes [f], Britta Galling [g], Ralitza Gueorguieva [h], Siegfried Kasper [i], Anatoly Kreinin [j], Jimmi Nielsen [k,l], René Ernst Nielsen [m,aa], Gary Remington [n], Eila Repo-Tiihonen [o], Christian Schmidt-Kraepelin [p,q], Saeed S. Shafti [r], Le Xiao [s], Christoph U. Correll [t,u,v,w,x], Robbert-Jan Verkes [y,z]

[a] *Pro Persona Mental Health Care, Expertise Center for Depression, Nijmeegsebaan 61, 6525 DX Nijmegen, the Netherlands*
[b] *Department for Health Evidence, Radboud University Medical Center, Nijmegen, the Netherlands*
[c] *Dimence Mental Healthcare, Deventer, the Netherlands*
[d] *Dokuz Eylul University Faculty of Medicine, Department of Psychiatry, Balçova, İzmir, Turkey*
[e] *Hacettepe University Faculty of Medicine, Department of Psychiatry, Sıhhiye, Ankara, Turkey*
[f] *Division of Psychiatry, Imperial College London, London, UK*
[g] *Department of Child and Adolescent Psychiatry and Psychotherapy, Centre for Integrative Psychiatry, School of Medicine, Kiel, Germany*
[h] *Department of Biostatistics, Yale School of Public Health, New Haven, USA*
[i] *Medical University of Vienna, Center for Brain Research, Department of Molecular Neuroscience, Vienna, Austria*
[j] *Ariel University, Ariel, Israel*
[k] *Department of Clinical Medicine, University of Copenhagen, Faculty of Health and Medical Sciences, Copenhagen, Denmark*
[l] *Mental health Centre Glostrup, Mental health service Capital Region Denmark, Copenhagen, Denmark*
[m] *Psychiatry, Aalborg University Hospital, Aalborg, Denmark*
[n] *Schizophrenia Division, Centre for Addiction and Mental Health (CAMH), University of Toronto, Toronto, Ontario, Canada*
[o] *University of Eastern Finland, Finland*
[p] *Kaiserswerther Diakonie, Florence-Nightingale-Hospital, Department of Psychiatry and Psychotherapy, Düsseldorf, Germany*
[q] *LVR-Clinic Düsseldorf, Department of Psychiatry and Psychotherapy, Medical Faculty, Heinrich-Heine-University, Duesseldorf, Germany*
[r] *University of Social Welfare and Rehabilitation Sciences, Tehran, Iran*
[s] *The National Clinical Research Center for Mental Disorders, Mood Disorders Center, Beijing Anding Hospital Capital Medical University, Beijing, China*
[t] *The Zucker Hillside Hospital, Department of Psychiatry, Northwell Health, Glen Oaks, NY, USA*
[u] *Donald and Barbara Zucker School of Medicine at Hofstra/Northwell, Department of Psychiatry and Molecular Medicine, Hempstead, NY, USA*
[v] *The Feinstein Institute for Medical Research, Center for Psychiatric Neuroscience, Northwell Health, New Hyde Park, NY, USA*
[w] *Charité – Universitätsmedizin Berlin, Department of Child and Adolescent Psychiatry, Berlin, Germany*
[x] *DZPG, German Center for Mental Health, Partner Site Berlin, Germany*
[y] *Department of Psychiatry, Radboud University Medical Center, Nijmegen, the Netherlands*
[z] *Donders Institute for Brain, Cognition, and Behaviour, Radboud University, Nijmegen, the Netherlands*
[aa] *Aalborg University, Department of Clinical Medicine, Aalborg, Denmark*

ARTICLE INFO

Keywords:
Antipsychotic polypharmacy
Efficacy
Tolerability
Effect moderators

ABSTRACT

Background: Antipsychotic polypharmacy (APP) is frequently prescribed for schizophrenia-spectrum disorders. Despite the inconsistent findings on efficacy, APP may be beneficial for subgroups of psychotic patients. This meta-analysis of individual patient data investigated moderators of efficacy and tolerability of APP in adult patients with schizophrenia-spectrum disorders.
Design: We searched PubMed, EMBASE, and the Cochrane Central Register of Randomized Trials until September 1, 2022, for randomized controlled trials comparing APP with antipsychotic monotherapy. We estimated the effects with a one-stage approach for patient-level moderators and a two-stage approach for study-level

* Corresponding author.
E-mail address: m.lochmannvanbennekom@propersona.nl (M.W.H. Lochmann van Bennekom).

https://doi.org/10.1016/j.schres.2024.07.035
Received 10 December 2023; Received in revised form 14 July 2024; Accepted 15 July 2024
Available online 13 August 2024

moderators, using (generalized) linear mixed-effects models. Primary outcome was treatment response, defined as a reduction of 25 % or more in the Positive and Negative Syndrome Scale (PANSS) score. Secondary outcomes were study discontinuation, and changes from baseline on the PANSS total score, its positive and negative symptom subscale scores, the Clinical Global Impressions Scale (CGI), and adverse effects.
Results: We obtained individual patient data from 10 studies (602 patients; 31 % of all possible patients) and included 599 patients in our analysis. A higher baseline PANSS total score increased the chance of a response to APP (OR = 1.41, 95 % CI 1.02; 1.94, $p = 0.037$ per 10-point increase in baseline PANSS total), mainly driven by baseline positive symptoms. The same applied to changes on the PANSS positive symptom subscale and the CGI severity scale. Extrapyramidal side effects increased significantly where first and second-generation antipsychotics were co-prescribed. Study discontinuation was comparable between both treatment arms.
Conclusions: APP was effective in severely psychotic patients with high baseline PANSS total scores and predominantly positive symptoms. This effect must be weighed against potential adverse effects.

1. Introduction

Schizophrenia and other psychotic disorders are severe, disruptive psychiatric disorders. Since decades, dysregulation of dopaminergic functioning has been postulated as a central explanatory model, assuming that striatal presynaptic hyperdopaminergia involving D_2 receptors underlies psychotic symptoms and cortical hypodopaminergia involving D_1 receptors underlies cognitive symptoms in schizophrenia, and reducing striatal hyperdopaminergia is a considered to be a key mechanism of action in treating positive symptoms of schizophrenia (Kaar et al., 2020; McCutcheon et al., 2020). All approved antipsychotic medications reduce dopaminergic transmission, generally by blocking 60–80 % of postsynaptic dopamine D_2 receptors in the striatal region of the brain (Kaar et al., 2020), although clozapine, with its evidence-based superior efficacy in treatment-resistant psychosis, has only modest affinity for the D_2 receptor. However, treatment-resistant symptoms persist in approximately 40 % of patients, despite multiple antipsychotic monotherapies, including clozapine (Diniz et al., 2023), prompting clinicians to consider alternative treatments, including antipsychotic polypharmacy (APP). APP is the concurrent use of at least two different antipsychotic medications for a patient and is frequently prescribed in approximately 20 % of patients with a psychotic disorder (Gallego et al., 2012). Clinicians' attitudes towards APP are heterogeneous (Ajayi and Arora, 2023; Chang and Kim, 2014; Correll et al., 2011; James et al., 2017; Kishimoto et al., 2013), but the main reason for prescribing APP is to treat refractory psychotic symptoms (Correll and Gallego, 2012; Sernyak and Rosenheck, 2004). This strategy has been based on various hypotheses: a pharmacodynamic hypothesis (combining antipsychotics with different receptor profiles, e.g., clozapine with an antipsychotic with high D_2 receptor affinity like sulpiride); a pharmacokinetic hypothesis (drug-drug interactions resulting in higher antipsychotic plasma levels); an acute-phase hypothesis (temporarily combining a sedating antipsychotic with a non-sedating antipsychotic in acutely exacerbated psychotic patients); and/or an adherence hypothesis (adding a second antipsychotic may mitigate the adverse effects of the primary medication) (Azorin and Simon, 2020; Guinart and Correll, 2020). However, the scientific evidence supporting these hypotheses is limited and the clinical evidence for the efficacy and safety of APP is controversial (Fleischhacker and Uchida, 2014; Galling et al., 2017; Lochmann van Bennekom et al., 2013). The American Psychiatric Association Practice Guideline for the Treatment of Patients With Schizophrenia and all international guidelines therefore advocate antipsychotic monotherapy (APM), including an adequate treatment trial of clozapine for treatment-resistant psychotic disorder, and state that there is only weak and inconsistent evidence for benefit with APP in treating patients with treatment-resistant schizophrenia (American Psychiatric Association, 2020; Correll et al., 2022). Nevertheless, there is some evidence from systematic reviews and meta-analyses that APP may be superior to APM in subgroups of patients, e.g., inpatients, patients with more hospitalizations, higher illness severity, and a low initial Global Assessment of Functioning (GAF) score, or in acutely exacerbated patients (Bighelli et al., 2022; Correll et al., 2009; Galling et al., 2017; Ortiz-Orendain et al., 2017; Paton et al., 2007; Taylor and Smith, 2009; Taylor et al., 2012; Wang et al., 2010). These subgroups may reflect the selection of patients with a more severe psychotic illness.

Compared with a traditional study-level meta-analysis, an individual patient data meta-analysis (IPDMA) offers important advantages, including granularity and statistical power, for the investigation of the impact of these potential effect moderators (Hannink et al., 2013). Here, we describe the results of the first IPDMA, to our knowledge, investigating the efficacy and safety of APP versus APM in adult patients with schizophrenia-spectrum disorders. Our aim was to determine whether there were patient and study characteristics that were associated with a better outcome with APP.

2. Materials and methods

2.1. Registration

The study was preregistered at PROSPERO (CRD42015009464).

2.2. Eligibility criteria

Eligible were individual patient data (IPD) from all double-blind randomized placebo-controlled trials (RCTs) comparing any combination of registered antipsychotic medications with APM in patients aged 18 years or older, at least 80 % of whom had a schizophrenia-spectrum diagnosis (i.e., schizophrenia, schizoaffective disorder, schizophreniform disorder, acute psychotic disorder, and psychotic disorder not otherwise specified). To quantify the outcome, studies had to assess the severity of positive and/or negative symptoms using a recognised rating scale. We also included studies combining antipsychotic medications to reduce side effects and studies converting APP to monotherapy. We set no language restrictions.

2.3. Identification and selection of studies

We performed a comprehensive computerized systematic literature search in PubMed, EMBASE, and the Cochrane Central Register of Randomized trials from their inception until September 1, 2022. In addition, we searched for eligible studies in reference lists, meeting abstracts, trial registers, and by word of mouth. For the search strategy see supplementary Table S1. Two researchers (MLvB, HG) independently reviewed title and abstract of the identified studies. Studies in Chinese were reviewed with the assistance of a Chinese translator (LX). Unless the study unanimously was excluded by title and abstract, the full text was reviewed and discussed for eligibility until consensus was reached.

2.4. Data collection, extraction, and standardization

We asked the first authors or other authors of the eligible studies to share with us their anonymized primary data. We sent monthly reminders, if there was no response after 12 weeks the trial was considered as unavailable. We extracted relevant data from the acquired datasets (supplementary Methods S1) and merged these in a new data set, that we analyzed. To make different psychopathology outcome scales mutually comparable, we applied established formulas to convert scores into total Positive and Negative Syndrome Scale (PANSS) scores, and PANSS positive symptom subscale and/or negative symptom subscale scores. We recalculated total antipsychotic end dose in olanzapine dose equivalents. For the formulas, see supplementary Table S2. Patients were considered dropout if this was registered as such in the IPD or (if such record was not available) no measurement had been recorded at the last visit.

2.5. Quality assessment

We checked the integrity of IPD by visual and digital inspection on completeness and consistency. Discrepancies were resolved with the original study authors. We used the Cochrane Risk of Bias 2 tool (RoB 2) to assess the risk of bias in individual studies (Higgins et al., 2021). We assessed potential selection bias by comparing key baseline characteristics of eligible studies from which we could and could not include IPD and attempted to assess the risk of publication bias.

2.6. Statistical analyses

The primary outcome was clinically relevant response, defined as a reduction of at least 25 % in total PANSS score, at primary study endpoint (see also supplementary Methods S2). This is an accepted, clinically meaningful, effect in patients with refractory psychotic disorder (Leucht et al., 2009; Leucht et al., 2006). Analyses of response were performed without discontinuation studies, because in these studies a difference in response does not adequately reflect the efficacy of APP versus APM. Secondary outcomes were the mean change from baseline on the PANSS total and the PANSS positive and negative symptom subscales, and changes on the Clinical Global Impressions scales for severity (CGI-S) and improvement (CGI-I). For the CGI-I subscale, we considered a rating of at least 'minimally improved' (ratings 1–3) as a clinically meaningful effect (Leucht et al., 2009; Leucht et al., 2006). When analyzing the CGI-I data, we also excluded discontinuation studies, for the aforementioned reason. Further outcomes were the frequency and severity of adverse effects and all-cause discontinuation. Potential moderators of effect that we investigated were:

1) Study aim (refractory psychotic symptoms, treating adverse effects, discontinuation of APP)
2) Study region (Europe, North America, Asia)
3) Illness stage (first episode, recurrent with acute exacerbation, chronic, refractory)
4) Illness duration
5) Illness severity at baseline
6) Setting (inpatient/outpatient)
7) Combinations of antipsychotic medications
8) Sex
9) Age

Analyses were performed on an intention-to-treat basis, as close as possible to 12 weeks after randomization, and with the last observation carried forward if needed. Effects of patient-level characteristics were analyzed with a centered one-stage approach to prevent ecological bias (Belias et al., 2019), with (generalized) linear mixed-effects models, after evaluation of possible nonlinear effects (supplementary Methods S3). Interaction coefficients of the potential moderators and subgroup results were summarized in forest plots, based on original units, and for the change in PANSS scores, also based on standardized mean differences (SMDs). Sensitivity analyses were performed to evaluate the robustness of the findings; 1) selecting the 1–99 % quantile of the continuous modifier values to exclude outliers and 2) for secondary (continuous) outcomes excluding discontinuation studies (that may include an enriched population of patients that responded to and tolerated APP). Secondary outcomes, except those on psychopathology and extrapyramidal side effects (EPS), were reported using descriptive statistics. We applied the PRISMA-IPD checklist (Stewart et al., 2015) to report the study (supplementary Table S3) and the GRADE methodology (Atkins et al., 2004) to rate the evidence (supplementary Table S4). Analyses were performed with the statistical packages SPSS (version 25), and R version 4.2.2. (R Core Team, 2022).

2.7. Medical ethical issues

Since this IPDMA study uses anonymous data from approved studies, no new institutional review board approval was required.

3. Results

3.1. Study selection

Our search yielded 2075 papers, from which we identified 31 non-overlapping eligible studies with 1957 patients. We obtained IPD from 10 studies (32 %), comprising 602 patients (31 %). IPD from 21 studies were not obtained: from four studies, authors were interested to participate but no data were received; from another four studies IPD were not available for various reasons; from two studies, authors refused cooperation; and from 11 studies we received no response at all (Supplementary Table S6). We excluded IPD from three patients (one from the study by Mossaheb and colleagues (Mossaheb et al., 2006) and two from the study by Kreinin and colleagues (Kreinin et al., 2006)) because of dropout before randomization, finally resulting in 599 patients. For the study flow diagram, see Fig. 1.

3.2. Study and patient characteristics

The included 10 IPD were related to nine 2-arm RCTs (Anil Yagcioglu et al., 2005; Barnes et al., 2018; Borlido et al., 2016; Gunduz-Bruce et al., 2013; Kreinin et al., 2006; Mossaheb et al., 2006; Nielsen et al., 2012a; Repo-Tiihonen et al., 2012; Shafti, 2009), and one 3-arm RCT (Schmidt-Kraepelin et al., 2022) published from 2005 to 2022. Seven studies investigated the effects of APP on refractory psychotic symptoms (one in acutely exacerbated, not clozapine-resistant patients (Schmidt-Kraepelin et al., 2022), six were in patients with treatment-resistant illness (Anil Yagcioglu et al., 2005; Barnes et al., 2018; Gunduz-Bruce et al., 2013; Mossaheb et al., 2006; Nielsen et al., 2012a; Shafti, 2009), two were 'discontinuation' studies, converting APP to APM (Borlido et al., 2016; Repo-Tiihonen et al., 2012), and one study aimed to reduce the side effects (hypersalivation) of the primary antipsychotic medication (Kreinin et al., 2006). Study endpoints ranged from 3 to 12 weeks. Two studies (Kreinin et al., 2006; Repo-Tiihonen et al., 2012) applied a cross-over design; we included only the first phase to avoid possible carry-over effects (Elbourne et al., 2002). All patients were diagnosed with schizophrenia or schizoaffective disorder. Seven studies (including one discontinuation study) investigated combinations of clozapine with a second antipsychotic (Anil Yagcioglu et al., 2005; Barnes et al., 2018; Gunduz-Bruce et al., 2013; Kreinin et al., 2006; Mossaheb et al., 2006; Nielsen et al., 2012a; Repo-Tiihonen et al., 2012), two studies investigated non-clozapine combinations (Schmidt-Kraepelin et al., 2022; Shafti, 2009), and one study investigated discontinuation of miscellaneous APP combinations (Borlido et al., 2016). To measure psychopathology, studies used the PANSS (Kay et al., 1987), the Brief Psychiatric Rating Scale (BPRS) (Overall and Gorham, 2016), the Scale for the Assessment of Positive Symptoms (SAPS) (Andreasen, 1990), the Scale for the Assessment of Negative Symptoms (SANS) (Andreasen, 1990), and the Clinical Global Impressions scale (CGI) (Guy, 1976). Side effects were measured mainly with the Simpson-Angus extrapyramidal side effects Scale (SAS) (Simpson and Angus, 1970), the Barnes Akathisia

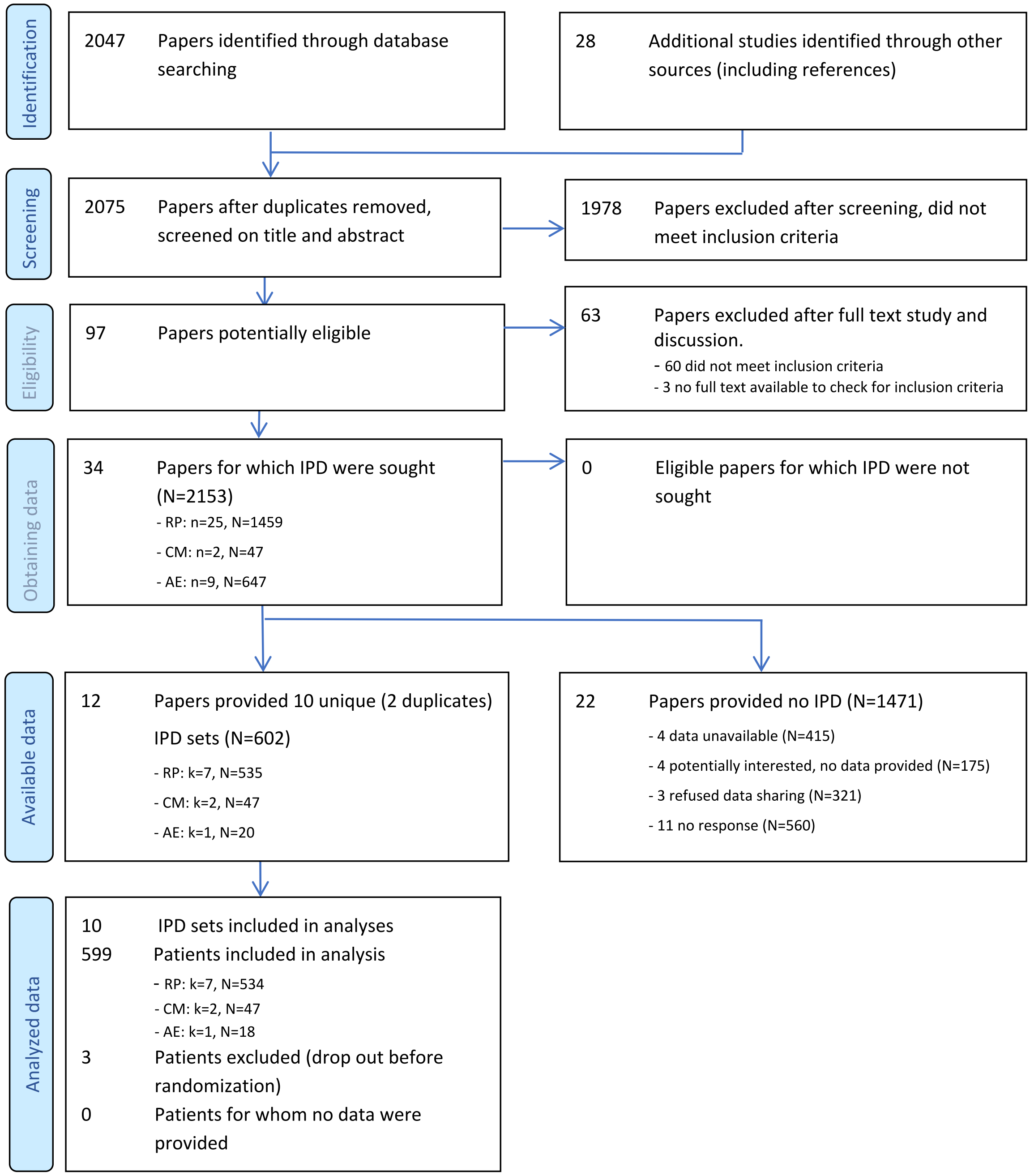

Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses of Individual Patient Data (PRISMA-IPD) Flow Diagram (final search: September 1, 2022)
Legend: AE = studies in patients with adverse effects, CM = studies converting antipsychotic polypharmacy to monotherapy, IPD = individual patient data, k = number of patient data sets, n = number of papers, N = number of participants, RP = studies in patients with refractory psychosis
The PRISMA IPD flow diagram © Reproduced with permission of the PRISMA IPD Group, which encourages sharing and reuse for non-commercial purposes. (Source: http://www.prisma-statement.org/Extensions/IndividualPatientData).

Rating Scale (BARS) (Barnes, 1989), and the Abnormal Involuntary Movement Scale (AIMS) (Guy, 1976). Inspection of the IPD revealed one unlikely PANSS total score that was corrected after consensus with the original researchers. The characteristics of the studies that provided the IPD are reported in Table 1. The baseline characteristics of the included patients are reported in supplementary Table S5. The characteristics of the 21 studies for which we could not obtain IPD are reported in supplementary Table S6. To assess the risk of selection bias we compared the characteristics of studies from which we could and could not obtain IPD. This showed that studies investigating the efficacy of APP were equally represented in approximately 30 % of studies, European studies were over-represented and North American studies were under-represented in our sample. Mainly, eligible studies investigated clozapine combinations for treatment refractory illness, that were equally represented (supplementary Table S7).

3.3. Risk of bias

The overall risk of bias of included studies according the RoB 2 tool was assessed as low in four studies (Anil Yagcioglu et al., 2005; Barnes et al., 2018; Borlido et al., 2016; Schmidt-Kraepelin et al., 2022), with some concerns in two studies (Nielsen et al., 2012a; Shafti, 2009), and a

Table 1
Characteristics of the included studies.

	Study	Study region (country)	n (I/C)	Intervention vs comparison	Mean (SD) antipsychotic dose in milligram OLA-eq (APP; APM)	Primary endpoint	Aim and study population	Psychopathology outcome scales	Other scales	Conclusion
Clozapine combinations										
1	(Anil Yagcioglu et al., 2005) / (Akdede et al., 2006)	West Asia (Turkey)	30 (16/14)	CLOZ + RIS vs. CLOZ	27.80 (5.77); 13.81 (3.23)	6w	Reducing refractory psychotic symptoms in in- and outpatients with schizophrenia, insufficiently responding to clozapine.	PANSS, CGI-S, CDS	SAS, AIMS, BAS, UKU, RAVLT, COWAT, DST, GAF, QoL-21	No significant benefit for APP with respect to psychopathology and cognitive functioning.
2	(Barnes et al., 2018)	Europe (UK)	68 (35/33)	CLOZ + AMI vs. CLOZ	26.61 (6.46); 12.46 (4.83)	12w	Reducing refractory psychotic symptoms in in- and outpatients with schizophrenia, insufficiently responding to clozapine (PANSS ≥ 80).	PANSS, CDS	SAS, AIMS, BARS,	No significant differences in therapeutic efficacy between both groups.
3	(Gunduz-Bruce et al., 2013)	North America (US)	28 (14/14)	CLOZ + PIM vs CLOZ	Unknown	12w	Reducing refractory psychotic symptoms in outpatients with schizophrenia or schizoaffective disorder, insufficiently responding to clozapine (BPRS ≥ 35).	SANS, BPRS, CGI-S, CGI-I	SAS, AIMS, RAVLT, COWAT, DST	No beneficial effect of APP.
4	(Kreinin et al., 2006)	West Asia (Israel)	20 (9/11)	CLOZ + AMI vs. CLOZ	Unknown	3w	Reducing hypersalivation in clozapine treated inpatients with schizophrenia.	PANSS, CGI-S, CGI-I	SAS, NHRS,	Significant improvement for APP group on the PANSS negative symptoms subscale, not on other subscales of the PANSS
5	(Mossaheb et al., 2006)	Europe (Austria)	10 (3/7)	CLOZ + HAL vs. CLOZ	19.75 (2.12); 15.00 (2.12)	10w	Reducing refractory psychotic symptoms in patients with schizophrenia, insufficiently responding to clozapine.	PANSS, CGI-S, CGI-I	SAS, AIMS, BARS	No significant difference in PANSS total scores between both groups.
6	(Nielsen et al., 2012a) / (Nielsen et al., 2012b)	Europe (Denmark)	50 (25/25)	CLOZ + SER vs. CLOZ	21.90 (4.44); 13.05 (5.93)	12w	Reducing refractory psychotic symptoms in patients with schizophrenia, insufficiently responding to clozapine.	PANSS, CGI-S, CGI-I	UKU, QoL-26, GAF	APP was not superior to monotherapy.
non-clozapine combinations										
7	(Schmidt-Kraepelin et al., 2022)	Europe (Germany)	321 (110/102/109)	OLA + AMI vs. OLA vs. AMI	28.28 (11.80); 14.13 (5.56); 16.99 (6.86)	8w	Reducing psychotic symptoms in inpatients with non-first episode schizophrenia or schizoaffective disorder (PANSS ≥ 70, at least two positive subscale items rated ≥ 4), excluding patients with a history of clozapine failure.	PANSS, CGI-S, CGI-I	SAS, DOTES, DISF-SR, SWN-S	AMI + OLA was significantly more effective than OLA monotherapy. No significant difference was observed between AMI + OLA and AMI monotherapy.
8	(Shafti, 2009)	West Asia (Iran)	28 (14/14)	OLA + FLU-DEC vs. OLA	Unknown	12w	Reducing psychotic symptoms in female inpatients with schizophrenia insufficiently	SANS, SAPS, CGI-S	SAS	APP was significantly more effective than monotherapy on SAPS and CGI-S outcomes.

(continued on next page)

Table 1 (*continued*)

	Study	Study region (country)	n (I/C)	Intervention vs comparison	Mean (SD) antipsychotic dose in milligram OLA-eq (APP; APM)	Primary endpoint	Aim and study population	Psychopathology outcome scales	Other scales	Conclusion
							responsive to olanzapine.			
Discontinuation studies										
9	(Borlido et al., 2016)	North America (Canada)	35 (17/18)	APP vs. APM	19.92 (9.88) median 18.34; 20.72 (16.62) median 14.71)	12w	Conversion to monotherapy in in- and outpatients with schizophrenia or schizoaffective disorder ≥ 30 days on APP	BPRS, CGI-S, CGI-I	SAS, AIMS, BARS	Almost 80 % could be safely transitioned from APP to APM with no clinical deterioration.
10	(Repo-Tiihonen et al., 2012)	Europe (Finland)	12 (5/7)	CLOZ + OLA vs. CLOZ	42.85 (8.28); 17,14 (6,52)	12w	Conversion to clozapine monotherapy in inpatients with schizophrenia treated with clozapine-olanzapine combination.	CGI-S, CGI-I	GAF	The clinical state of patients who were on OLA + CLOZ therapy was not affected by discontinuation of olanzapine.

Legend: AIMS = Abnormal Involuntary Movement Scale, AMI = amisulpride, APP = antipsychotic polypharmacy, ARI = aripiprazole, BARS=Barnes Akathisia Rating Scale, BPRS=Brief Psychiatric Rating Scale, CDS=Calgary Depression Scale, CGI-I=Clinical Global Impressions Improvement scale, CGI-S=Clinical Global Impressions Severity scale, CLOZ = clozapine, COWAT = Controlled Word Association Test, DISF-SR = Derogatis Interview for Sexual Functioning–Self Reporting, DOTES=Dosage Record and Treatment Emergent Symptom Scale, DST = Digit Span Test, FLU-DEC = fluphenazine decanoate, GAF = Global Assessment of Functioning, HAL = haloperidol, n(I/C) = total number of patients (number allocated to intervention arm/number allocated to control arm), NHRS=Nocturnal Hypersalivation Rating Scale, OLA = olanzapine, OLA-eq = olanzapine equivalent, PANSS=Positive and Negative Syndrome Scale, p.e. = primary endpoint, PIM = pimozide, QoL-21 = Quality of Life scale 21 items, QoL-26 = Quality of Life scale 26 items, RAVLT = Rey Auditory Verbal Learning Test, RIS = risperidone, SANS=Scale for the Assessment of Negative Symptoms, SAPS=Scale for the Assessment of Positive Symptoms, SAS=Simpson-Angus extrapyramidal side effects Scale, SER = sertindole, SWN-S=Subjective Wellbeing under Neuroleptics Scale–Short form, UKU=Udvalg for Kliniske Undersøgelser side effect rating scale.

high risk in four studies (Gunduz-Bruce et al., 2013; Kreinin et al., 2006; Mossaheb et al., 2006; Repo-Tiihonen et al., 2012) (supplementary Table S8). The number of studies was too small to evaluate potential publication bias.

3.4. Efficacy of APP

3.4.1. Is APP more efficacious than APM in terms of clinically relevant response (i.e., ≥25 % reduction) on PANSS outcomes?

3.4.1.1. Overall. The weighted aggregated data of seven included datasets revealed that 38 % of patients on APP and 30 % on APM met the criterion for a clinically relevant response on the PANSS total score. There was no statistically significant difference between the two groups (odds ratio [OR] = 1.37 for APP versus APM, 95 % confidence interval [CI] 0.80; 2.33, p = 0.199).

3.4.1.2. Study-level characteristics. There were insufficient studies per characteristic to draw any robust conclusions about possible interactions between study-level characteristics (study aim, study region or stage of illness) and treatment group on response to APP.

3.4.1.3. Patient-level characteristics. The odds of a response to APP versus APM on the PANSS total outcome increased by 41 % per 10-point increase in baseline PANSS total score (OR = 1.41, 95 % CI 1.02; 1.94, p = 0.037), which appeared robust in sensitivity analyses. The odds of a response to APP on the PANSS positive symptom subscale outcome doubled per 5-point increase in baseline PANSS positive symptom subscale score (OR = 2.02, 95 % CI 1.30; 3.12, p = 0.002). The odds of response to APP on the PANSS negative symptom subscale outcome increased by 45 % per 10-point increase in baseline PANSS total score (OR = 1.45, 95 % CI 1.04; 2.02, p = 0.03). Response was not notably moderated by any of the other investigated characteristics. There was no significantly beneficial effect for any of the investigated antipsychotic combinations on response.

For an overview of the results on response see Fig. 2a.

3.4.2. Is APP more efficacious than APM in terms of change on PANSS or CGI?

3.4.2.1. Overall. Overall, there was no significant difference in mean change on PANSS total score between the two groups (mean difference [MD] = −1.08 for APP versus APM, 95 % CI -5.69; 3.53, p = 0.596).

3.4.2.2. Study-level characteristics. There were insufficient studies per characteristic to draw any robust conclusions about a possible interaction between study-level characteristics and treatment group on change on the PANSS scores, or on the CGI.

3.4.2.3. Patient-level characteristics. Reduction of the PANSS total outcome score was significantly greater for APP versus APM in patients with higher baseline PANSS total scores (MD = -0.21, 95 % CI -3.99; −0.20, p = 0.031 per 10-point increase on baseline PANSS total score). Reduction of the PANSS total outcome score was also significantly greater for APP versus APM in patients with higher baseline PANSS positive symptom subscale scores (MD = -2.77, 95 % CI -5.45; −0.09, p = 0.043 per 5-point increase on baseline PANSS positive subscale score). In sensitivity analyses, both interactions were no longer significant.

Reduction of the PANSS positive symptom subscale outcome score was significantly greater for APP versus APM, both in patients with higher baseline PANSS total scores (MD = -0.66, 95 % CI -1.28; −0.004, p = 0.038 per 10 points increase in baseline PANSS total score) and in patients with higher baseline PANSS positive symptom subscale scores (MD = -1.21, 95 % CI -2.10; −0.32, p = 0.008 per 5-point increase in baseline PANSS positive subscale score), but not in patients with higher baseline PANSS negative symptom subscale scores. These findings

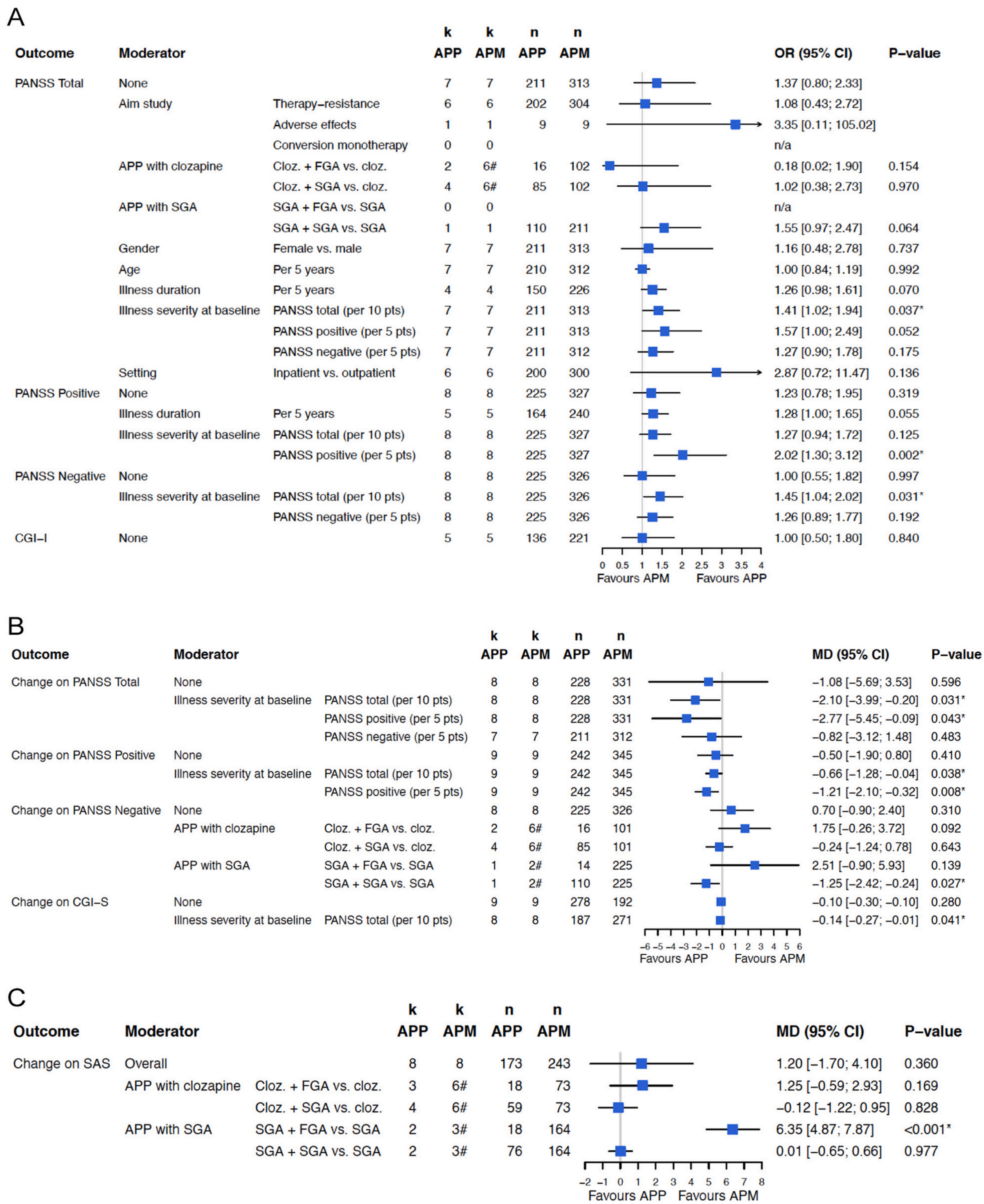

Fig. 2. Forest plots representing interaction of moderators for efficacy and tolerability of APP compared with APM on PANSS total, PANSS positive and negative subscales, CGI, and SAS outcomes

2a. Main moderators for response (i.e., ≥25 % improvement on PANSS, or at least minimally improvement on CGI-I)

2b. Main moderators for change from baseline on PANSS and CGI-S.

2c. Main moderators for extrapyramidal side effects on SAS

Legend: APM = antipsychotic monotherapy, APP = antipsychotic polypharmacy, CGI-I=Clinical Global Impressions Improvement scale, CGI-S=Clinical Global Impressions Severity scale, CI = confidence interval, Cloz. = clozapine, FGA = first generation antipsychotic, k = number of datasets, MD = mean difference, n = number of patients, OR = odds ratio, PANSS=Positive and Negative Syndrome Scale, SAS=Simpson-Angus extrapyramidal side effects Scale, SGA = second generation antipsychotic.

* = *p* value <0.05

= the number of studies in APP and APM group is not equal in the analyses of combinations of antipsychotics. This is because the placebo condition included patients from the APM group of the combination study in question, as well as patients from the APM group of other combination studies.

appeared robust in sensitivity analyses.

Reduction of the PANSS negative symptom subscale outcome score was significantly greater with combinations of two SGAs (MD = -1.25, 95 % CI -2.42; −0.24, $p = 0.027$), but was not significantly moderated by any of the other investigated characteristics.

Reduction of the CGI-S outcome score was also significantly greater for patients with higher baseline PANSS total scores (MD = -0.14, 95 % CI -0.27; −0.01, $p = 0.041$ per 10-point increase in baseline PANSS total score), which appeared robust in sensitivity analyses. There were no notable interactions between any of the investigated characteristics on treatment outcome on the CGI-I.

For an overview of the results on change from baseline, see the forest plot in Fig. 2b.

For both APP and APM, the association between baseline illness severity on PANSS total score and the probability of respectively response and change from baseline on the PANSS total outcome score is visualized in Fig. 3. We found similar results for the standardized differences of means. For the forest plots of all the investigated effect moderators, see supplementary Figs. S1a – d.

3.5. Tolerability and other secondary outcomes

3.5.1. Tolerability measured with the SAS

There was no overall difference between the two groups in EPS, determined with the SAS. However, EPS were significantly more frequent with FGA-SGA combinations (MD = 6.35, 95 % CI 4.87; 7.87, $p < 0.001$). None of the other investigated moderators interacted significantly with treatment on the SAS. For an overview, see Fig. 2c.

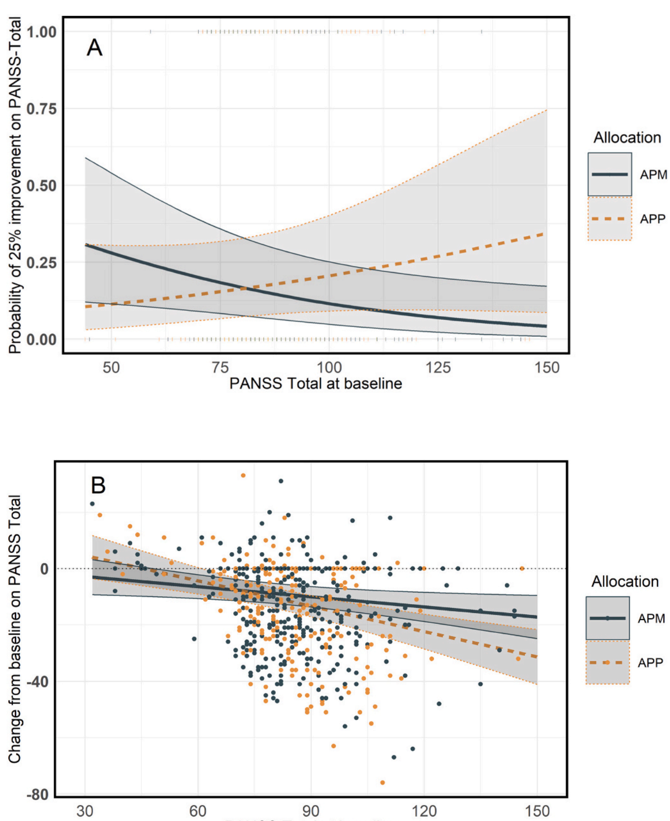

Fig. 3. Association between the level of baseline PANSS total score and the predicted PANSS total outcome
3a. Probability of ≥25 % improvement on PANSS total outcome
3b. Probability of change from baseline on PANSS total outcome
Legend: APM = antipsychotic monotherapy, APP = antipsychotic polypharmacy, PANSS = Positive and Negative Syndrome Scale.

3.5.2. Tolerability measured with the AIMS, BARS and study discontinuation

The descriptive results of study discontinuation and adverse effects outcome scales available applied in at least four datasets; these are presented in Table 2. Discontinuation in the APP group (21 %) was lower than in the APM group (28 %). Outcomes on the AIMS and BARS scores were in the lower range in both groups. Due to limited/lacking data we cannot report results for cognition, residual mood symptoms, quality of life, and cost outcomes.

4. Discussion

We analyzed IPD on 599 patients, derived from 10 studies that were published before September 1, 2022, representing 32 % of all eligible studies and 31 % of all eligible patients. We found that for patients with higher baseline PANSS total scores, the odds to achieve response on the PANSS total outcome score were significantly higher for those receiving APP rather than APM. Analysis of the PANSS subscales outcomes revealed that this effect was more attributable to improvement in positive than in negative symptoms. The probability of a superior response to APP increased in patients with baseline PANSS total scores (range: 30–210) in the higher levels (in this sample approximately above 110): with every 10-point increase in baseline PANSS total score, the superiority of APP increased with approximately 2 points more reduction in PANSS total outcome score after 3–12 weeks. We could not identify specific combinations of antipsychotic medications that were more effective in reducing PANSS total or positive symptoms subscale outcome scores. Regarding the reduction of negative symptoms, combinations of two SGAs were superior to combinations involving an FGA and/or clozapine. None of the other investigated characteristics were associated with a better outcome for APP.

Pharmacotherapy with antipsychotic medication is an important cornerstone in the treatment of patients with psychotic disorders. In the absence of sufficiently explanatory etiological or neurobiological models for psychosis, the classical hypothesis is that antipsychotics reduce postsynaptic dopamine transmission in the mesolimbic area of the brain by blocking the dopamine D_2 receptor. However, at therapeutic doses, an optimum of 65 % D_2 receptor blocking for maximum antipsychotic effect is reached, implicating that a higher antipsychotic load targeting D_2 receptors will not result in greater efficacy (Kapur et al., 2000). Clozapine has superior antipsychotic efficacy in treating patients with refractory psychotic symptoms and has a complex receptor profile with only low affinity for the D2 receptor, suggesting that other receptors may play an essential role in efficacy of antipsychotics on positive and negative symptoms, such as modulation of the D_3 and serotonergic receptors (Hjorth, 2021). Combining antipsychotic

Table 2
Descriptive statistics of secondary outcomes reported in ≥4 IPD.

	APP		APM	
Variable	Frequency (%)		Frequency (%)	
All-cause discontinuation (k = 10, $n = 599$)	52 (21.1 %)		99 (28.1 %)	
Variable	Baseline (SD)	Change from baseline (SD)	Baseline (SD)	Change from baseline (SD)
Adverse effects				
- AIMS (range: 0–28) (k = 5; $n = 97$)	1.33 (2.60)	0.18 (1.09)	1.69 (3.67)	0.11 (1.42)
- BARS (range: 0–9) (k = 4; $n = 130$)	0.69 (1.15)	0.17 (0.92)	1.03 (1.73)	0.28 (2.29)

Legend: AIMS = Abnormal Involuntary Movement Scale, APM = antipsychotic monotherapy, APP = antipsychotic polypharmacy, BARS = Barnes Akathisia Rating Scale, IPD = individual patient data, k = number of patient data sets, n = number of participants, SD = standard deviation.

medications with different receptor profiles may to some extent explain any superiority of APP, although it cannot explain the difference in efficacy we found between more and less severely psychotic patients.

A better response on APP compared with APM in those patients with a more severe psychotic illness is consistent with a previous meta-analysis where the addition of a partial D_2 agonist to a full antagonist was associated with superiority of APP versus APM in open-label and low-quality (however, not in double blind randomized controlled) trials (Galling et al., 2017). This finding cannot be explained by 'regression to the mean', since we included only placebo-controlled RCTs with balanced baseline illness severity in both treatment groups. Our finding that APP is beneficial for the most severely ill patients is also consistent with studies that have found discontinuation of APP to be less successful in more chronically and severely ill patients (Borlido et al., 2016; Constantine et al., 2015).

Our finding of a beneficial effect of combined SGAs on negative symptoms may be in line with previous findings that augmentation of D_2 antagonists with a partial D_2 agonist was associated with significantly reduced negative symptoms (Galling et al., 2017). Our study did not allow to identify specific effective combinations.

A reduction of 25 % in the PANSS total score reflects a reduction of the CGI-S by one severity step while a 15-point reduction in the PANSS total score approximately corresponds to minimal improvement on the CGI-I and a change on the CGI-S by one point (Leucht et al., 2006). Thus, the magnitude of the effect in our sample was modest. Efficacy was observed with the PANSS and the CGI-S, but not with the CGI-I scale. Clinically this implies that APP treatment should be carefully assessed for any added benefit, which should be weighed against the presence of and potential risk of adverse effects.

We found a significant increase in EPS in those patients treated with both an FGA and an SGA. There was no notable difference in all-cause discontinuation between both groups. Because of limited/lacking data, we could not analyze cognitive functioning, mood symptoms, quality of life, or medical health costs.

4.1. Strengths and limitations of this study

To the best of our knowledge, this is the first IPDMA to investigate the characteristics of patients with schizophrenia-spectrum disorders that might benefit from APP. A major strength of the current study is that it allowed effect moderators to be investigated with more statistical power than is possible in study-level meta-analyses. The studies included varied in their aims, and comprised patients in various stages of their illness, which enhances the generalizability of the findings.

Several limitations should be considered. First, we were only able to obtain 31 % of potentially eligible IPD, which is a relatively low retrieval rate (Wang et al., 2021), and may limit the generalizability of our findings. However, our overall comparison of studies from which we could and could not obtain IPD demonstrated that the former were a representative reflection of all studies conducted in this area, which are dominated by studies investigating the efficacy of APP in patients with treatment refractory psychotic illness (67–70 %), focusing mainly on combinations including clozapine (67 %) (supplementary Table S8). Secondly, superiority of APP in more severely ill patients could be explained by the higher mean total antipsychotic dose in the APP group compared with the APM group (26.8 resp. 15.3 mg olanzapine equivalents). Unfortunately, the design of the relevant studies did not allow us to unravel this explanation, which would call for differently designed studies with increasing doses in the monotherapy arm (Azorin and Simon, 2020). However, a previous systematic review did not find any therapeutic advantage for higher antipsychotic dosing (Samara et al., 2018). Thirdly, the included studies represented only a proportion of the possible antipsychotic combinations and were analyzed in broad categories (FGA, SGA, and clozapine combinations), impairing robust conclusions about the effectiveness of specific combinations of antipsychotic medications in particular patient subgroups.

5. Conclusions

The efficacy of APP versus APM in patients with schizophrenia-spectrum disorders depends on the severity of the disorder: APP is more effective for patients with high PANSS total scores, driven mostly by the positive symptoms. In those with less severe refractory illness and predominantly negative symptoms, APP appears to be no more effective than APM. Any incremental benefit for APP versus APM was modest and should be individually weighed against side-effect burden, especially when combining an FGA with an SGA. The findings of this meta-analysis may be helpful in the future revision of guidelines and for those clinicians making treatment decisions for severely ill patients with schizophrenia-spectrum disorders. More research is needed to identify which combinations of antipsychotics are favorable, and to determine the impact of APP on important non-psychopathological parameters, like functional outcome, quality of life, and cost-effectiveness.

CRediT authorship contribution statement

Marc W.H. Lochmann van Bennekom: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joanna IntHout:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Harm J. Gijsman:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Berna B.K. Akdede:** Writing – review & editing. **A. Elif Anıl Yağcıoğlu:** Writing – review & editing. **Thomas R.E. Barnes:** Writing – review & editing. **Britta Galling:** Writing – review & editing. **Ralitza Gueorguieva:** Writing – review & editing. **Siegfried Kasper:** Writing – review & editing. **Anatoly Kreinin:** Writing – review & editing. **Jimmi Nielsen:** Writing – review & editing. **René Ernst Nielsen:** Writing – review & editing. **Gary Remington:** Writing – review & editing. **Eila Repo-Tiihonen:** Writing – review & editing. **Christian Schmidt-Kraepelin:** Writing – review & editing. **Saeed S. Shafti:** Writing – review & editing. **Le Xiao:** Writing – review & editing. **Christoph U. Correll:** Writing – review & editing, Conceptualization. **Robbert-Jan Verkes:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

Dr. Anıl Yağcıoğlu has served as a speaker/advisory board member for Janssen, Abdi İbrahim Otsuka, Boehringer Ingelheim, Santa Farma, Nobel and has been an investigator for Janssen, Boehringer Ingelheim. Dr. Kasper served in the past 3 years as a consultant or on advisory boards for Angelini, Biogen, Boehringer, Esai, Janssen, IQVIA, Mylan, Recordati, Rovi, Sage and Schwabe; and he has served on speakers bureaus for Angelini, Aspen Farmaceutica S.A., Biogen, Janssen, Recordati, Schwabe, Servier, Sothema, and Sun Pharma. Dr. R.E. Nielsen has received funding for research or has been an investigator for H. Lundbeck, Otsuka Pharmaceuticals, Compass, Boehringer Ingelheim and Janssen-Cilag. Furthermore, REN has received speakers fee from Bristol-Meyers Squibb, Astra Zeneca, Janssen-Cilag, Lundbeck, Servier, Otsuka Pharmaceuticals, Teva and Eli Lilly. Dr. Remington has received research support from the Canadian Institutes of Health Research (CIHR), University of Toronto, and HLS Therapeutics Inc. Dr. Schmidt-Kaepelin is patent holder of patent No.: 102020106962, and has received speakers honoraria by Boehringer Ingelheim. Dr. Correll has been a consultant and/or advisor to or has received honoraria from: AbbVie, Acadia, Alkermes, Allergan, Angelini, Aristo, Biogen, Boehringer-Ingelheim, Cardio Diagnostics, Cerevel, CNX Therapeutics, Compass Pathways, Darnitsa, Denovo, Gedeon Richter, Hikma, Holmusk, IntraCellular Therapies, Jamjoom Pharma, Janssen/J&J, Karuna, LB Pharma, Lundbeck, MedAvante-ProPhase, MedInCell, Merck, Mindpax, Mitsubishi Tanabe Pharma, Mylan, Neurocrine, Neurelis, Newron, Noven, Novo Nordisk, Otsuka, Pharmabrain, PPD Biotech, Recordati,

Relmada, Reviva, Rovi, Sage, Seqirus, SK Life Science, Sumitomo Pharma America, Sunovion, Sun Pharma, Supernus, Takeda, Teva, Tolmar, Vertex, and Viatris. He provided expert testimony for Janssen and Otsuka. He served on a Data Safety Monitoring Board for Compass Pathways, Denovo, Lundbeck, Relmada, Reviva, Rovi, Supernus, and Teva. He has received grant support from Janssen and Takeda. He received royalties from UpToDate and is also a stock option holder of Cardio Diagnostics, Kuleon Biosciences, LB Pharma, Mindpax, and Quantic. Lochmann van Bennekom, IntHout, Gijsman, Akdede, Barnes, Galling, Gueorguieva, Kreinin, J. Nielsen, Repo-Tiihonen, Shafti, Xiao and Verkes reported no conflicts of interest.

Acknowledgements

Prof. dr. M. Rovers (Radboudumc, Nijmegen, the Netherlands) made valuable contributions to the design of the study.

Prof. dr. F. Zitman (Leiden UMC, Leiden, the Netherlands) also contributed to the design of the study. He passed away in the course of the present study before this paper was finished. We are most grateful for his contributions to this study and psychiatry in general.

We thank all the members of the COMBINE study group (a list of all the investigators can be found in the supplementary appendix of the original publication PMID: 35276079) for their contribution, enabling the inclusion of a large number of IPD in our study, as well as all other co-authors of the other original studies, who made it possible to prepare this paper.

Funding

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.schres.2024.07.035.

References

Ajayi, S.U., Arora, P., 2023. A survey study on clinicians' rationale and attitude towards the prescription of antipsychotic polypharmacy in the East Perth metropolitan area in Western Australia. Cureus 15 (4), e37234.

Akdede, B.B., Anil Yagcioglu, A.E., Alptekin, K., Turgut, T.I., Tumuklu, M., Yazici, M.K., Jayathilake, K., Tunca, Z., Gogus, A., Meltzer, H.Y., 2006. A double-blind study of combination of clozapine with risperidone in patients with schizophrenia: effects on cognition. J. Clin. Psychiatry 67 (12), 1912–1919.

American Psychiatric Association, 2020. The American Psychiatric Association Practice Guideline for the Treatment of Patients with Schizophrenia, Third edition. American Psychiatric Association, Washington, DC.

Andreasen, N.C., 1990. Methods for assessing positive and negative symptoms. Mod. Probl. Pharmacopsychiatry 24, 73–88.

Anil Yagcioglu, A.E., Kivircik Akdede, B.B., Turgut, T.I., Tumuklu, M., Yazici, M.K., Alptekin, K., Ertugrul, A., Jayathilake, K., Gogus, A., Tunca, Z., Meltzer, H.Y., 2005. A double-blind controlled study of adjunctive treatment with risperidone in schizophrenic patients partially responsive to clozapine: efficacy and safety. J. Clin. Psychiatry 66 (1), 63–72.

Atkins, D., Eccles, M., Flottorp, S., Guyatt, G.H., Henry, D., Hill, S., Liberati, A., O'Connell, D., Oxman, A.D., Phillips, B., Schunemann, H., Edejer, T.T., Vist, G.E., Williams Jr., J.W., Group, G.W, 2004. Systems for grading the quality of evidence and the strength of recommendations I: critical appraisal of existing approaches the GRADE working group. BMC Health Serv. Res. 4 (1), 38.

Azorin, J.M., Simon, N., 2020. Antipsychotic polypharmacy in schizophrenia: evolving evidence and rationale. Expert Opinion Drug Metab. Toxicol. 16 (12), 1175–1186.

Barnes, T.R., 1989. A rating scale for drug-induced akathisia. Br. J. Psychiatry 154, 672–676.

Barnes, T.R.E., Leeson, V., Paton, C., Marston, L., Osborn, D.P., Kumar, R., Keown, P., Zafar, R., Iqbal, K., Singh, V., Fridrich, P., Fitzgerald, Z., Bagalkote, H., Haddad, P. M., Husni, M., Amos, T., 2018. Amisulpride augmentation of clozapine for treatment-refractory schizophrenia: a double-blind, placebo-controlled trial. Ther. Adv. Psychopharmacol. 8 (7), 185–197.

Belias, M., Rovers, M.M., Reitsma, J.B., Debray, T.P.A., IntHout, J., 2019. Statistical approaches to identify subgroups in meta-analysis of individual participant data: a simulation study. BMC Med. Res. Methodol. 19 (1), 183.

Bighelli, I., Rodolico, A., Siafis, S., Samara, M.T., Hansen, W.P., Salomone, S., Aguglia, E., Cutrufelli, P., Bauer, I., Baeckers, L., Leucht, S., 2022. Antipsychotic polypharmacy reduction versus polypharmacy continuation for people with schizophrenia. Cochrane Database Syst. Rev. 8 (8), CD014383.

Borlido, C., Remington, G., Graff-Guerrero, A., Arenovich, T., Hazra, M., Wong, A., Daskalakis, Z.J., Mamo, D.C., 2016. Switching from 2 antipsychotics to 1 antipsychotic in schizophrenia: a randomized, double-blind, placebo-controlled study. J. Clin. Psychiatry 77 (1), e14–e20.

Chang, J., Kim, B., 2014. A survey study of the satisfaction and attitude of the Korean psychiatrists toward antipsychotic polypharmacy. Ann. Clin. Psychiatry 26 (1), 64–69.

Constantine, R.J., Andel, R., McPherson, M., Tandon, R., 2015. The risks and benefits of switching patients with schizophrenia or schizoaffective disorder from two to one antipsychotic medication: a randomized controlled trial. Schizophr. Res. 166 (1–3), 194–200.

Correll, C.U., Gallego, J.A., 2012. Antipsychotic polypharmacy: a comprehensive evaluation of relevant correlates of a long-standing clinical practice. Psychiatr. Clin. North Am. 35 (3), 661–681.

Correll, C.U., Rummel-Kluge, C., Corves, C., Kane, J.M., Leucht, S., 2009. Antipsychotic combinations vs monotherapy in schizophrenia: a meta-analysis of randomized controlled trials. Schizophr. Bull. 35 (2), 443–457.

Correll, C.U., Shaikh, L., Gallego, J.A., Nachbar, J., Olshanskiy, V., Kishimoto, T., Kane, J.M., 2011. Antipsychotic polypharmacy: a survey study of prescriber attitudes, knowledge and behavior. Schizophr. Res. 131 (1–3), 58–62.

Correll, C.U., Martin, A., Patel, C., Benson, C., Goulding, R., Kern-Sliwa, J., Joshi, K., Schiller, E., Kim, E., 2022. Systematic literature review of schizophrenia clinical practice guidelines on acute and maintenance management with antipsychotics. Schizophrenia (Heidelb) 8 (1), 5.

Diniz, E., Fonseca, L., Rocha, D., Trevizol, A., Cerqueira, R., Ortiz, B., Brunoni, A.R., Bressan, R., Correll, C.U., Gadelha, A., 2023. Treatment resistance in schizophrenia: a meta-analysis of prevalence and correlates. Braz. J. Psychiatry 45 (5), 448–458.

Elbourne, D.R., Altman, D.G., Higgins, J.P., Curtin, F., Worthington, H.V., Vail, A., 2002. Meta-analyses involving cross-over trials: methodological issues. Int. J. Epidemiol. 31 (1), 140–149.

Fleischhacker, W.W., Uchida, H., 2014. Critical review of antipsychotic polypharmacy in the treatment of schizophrenia. Int. J. Neuropsychopharmacol. 17 (7), 1083–1093.

Gallego, J.A., Bonetti, J., Zhang, J., Kane, J.M., Correll, C.U., 2012. Prevalence and correlates of antipsychotic polypharmacy: a systematic review and meta-regression of global and regional trends from the 1970s to 2009. Schizophr. Res. 138 (1), 18–28.

Galling, B., Roldan, A., Hagi, K., Rietschel, L., Walyzada, F., Zheng, W., Cao, X.L., Xiang, Y.T., Zink, M., Kane, J.M., Nielsen, J., Leucht, S., Correll, C.U., 2017. Antipsychotic augmentation vs. monotherapy in schizophrenia: systematic review, meta-analysis and meta-regression analysis. World Psychiatry 16 (1), 77–89.

Guinart, D., Correll, C.U., 2020. Antipsychotic polypharmacy in schizophrenia: why not? J. Clin. Psychiatry 81 (3).

Gunduz-Bruce, H., Oliver, S., Gueorguieva, R., Forselius-Bielen, K., D'Souza, D.C., Zimolo, Z., Tek, C., Kaliora, S., Ray, S., Petrides, G., 2013. Efficacy of pimozide augmentation for clozapine partial responders with schizophrenia. Schizophr. Res. 143 (2–3), 344–347.

Guy, W.E., 1976. ECDEU Assessment Manual for Psychopharmacology: Revised. US Department of Health, Education, and Welfare, Public Health Service Alcohol, Drug Abuse, and Mental Health Administration, NIMH Psychopharmacology Research Branch, Division of Extramural Research Programs, Rockville, MD.

Hannink, G., Gooszen, H.G., van Laarhoven, C.J., Rovers, M.M., 2013. A systematic review of individual patient data meta-analyses on surgical interventions. Syst. Rev. 2, 52.

Higgins .J., J.P.T.S., Page, M.J., Elbers, R.G., Sterne, J.A.C., 2021. Assessing risk of bias in a randomized trial. In: Higgins JPT, T.J., Chandler, J., Cumpston, M., Li, T., Page, M.J., Welch, V.A. (Eds.), Cochrane Handbook for Systematic Reviews of Interventions version 6.2 (updated February 2021). Cochrane Collaboration. Available from. www.training.cochrane.org/handbook.

Hjorth, S., 2021. The more, the merrier...? Antipsychotic polypharmacy treatment strategies in schizophrenia from a pharmacology perspective. Front. Psychiatry 12, 760181.

James, B.O., Omoaregba, J.O., Raji, S.O., Imishue, O.E., Okonoda, K.M., Nyamali, Y.I., Famuyiwa, P.A., Correll, C.U., 2017. Attitudes towards and rationale for antipsychotic polypharmacy among psychiatrists in Nigeria: characteristics associated with high reported antipsychotic polypharmacy. Psychiatry Res. 248, 134–139.

Kaar, S.J., Natesan, S., McCutcheon, R., Howes, O.D., 2020. Antipsychotics: mechanisms underlying clinical response and side-effects and novel treatment approaches based on pathophysiology. Neuropharmacology 172, 107704.

Kapur, S., Zipursky, R., Jones, C., Remington, G., Houle, S., 2000. Relationship between dopamine D(2) occupancy, clinical response, and side effects: a double-blind PET study of first-episode schizophrenia. Am. J. Psychiatry 157 (4), 514–520.

Kay, S.R., Fiszbein, A., Opler, L.A., 1987. The positive and negative syndrome scale (PANSS) for schizophrenia. Schizophr. Bull. 13 (2), 261–276.

Kishimoto, T., Watanabe, K., Uchida, H., Mimura, M., Kane, J.M., Correll, C.U., 2013. Antipsychotic polypharmacy: a Japanese survey of prescribers' attitudes and rationales. Psychiatry Res. 209 (3), 406–411.

Kreinin, A., Novitski, D., Weizman, A., 2006. Amisulpride treatment of clozapine-induced hypersalivation in schizophrenia patients: a randomized, double-blind, placebo-controlled cross-over study. Int. Clin. Psychopharmacol. 21 (2), 99–103.

Leucht, S., Kane, J.M., Etschel, E., Kissling, W., Hamann, J., Engel, R.R., 2006. Linking the PANSS, BPRS, and CGI: clinical implications. Neuropsychopharmacology 31 (10), 2318–2325.

Leucht, S., Davis, J.M., Engel, R.R., Kissling, W., Kane, J.M., 2009. Definitions of response and remission in schizophrenia: recommendations for their use and their presentation. Acta Psychiatr. Scand. Supplementum(438), 7–14.
Lochmann van Bennekom, M.W., Gijsman, H.J., Zitman, F.G., 2013. Antipsychotic polypharmacy in psychotic disorders: a critical review of neurobiology, efficacy, tolerability and cost effectiveness. J. Psychopharmacol. 27 (4), 327–336.
McCutcheon, R.A., Krystal, J.H., Howes, O.D., 2020. Dopamine and glutamate in schizophrenia: biology, symptoms and treatment. World Psychiatry 19 (1), 15–33.
Mossaheb, N., Sacher, J., Wiesegger, G., Klein, N., Spindelegger, C., Asenbaum, S., Dudczak, R., Kasper, S., 2006. Haloperidol in combination with clozapine in treatment-refractory patients with schizophrenia. Eur. Neuropsychopharmacol. 16 (Suppl. 4), S416.
Nielsen, J., Emborg, C., Gydesen, S., Dybbro, J., Aagaard, J., Haderup, K., Glyngdal, P., Fabricius, S., Thode, D., Lublin, H., Andersen, T., Damkier, P., Taylor, D., 2012a. Augmenting clozapine with sertindole: a double-blind, randomized, placebo-controlled study. J. Clin. Psychopharmacol. 32 (2), 173–178.
Nielsen, R.E., Levander, S., Thode, D., Nielsen, J., 2012b. Effects of sertindole on cognition in clozapine-treated schizophrenia patients. Acta Psychiatr. Scand. 126 (1), 31–39.
Ortiz-Orendain, J., Castiello-de Obeso, S., Colunga-Lozano, L.E., Hu, Y., Maayan, N., Adams, C.E., 2017. Antipsychotic combinations for schizophrenia. Cochrane Database Syst. Rev. 6 (6), CD009005.
Overall, J.E., Gorham, D.R., 2016. The brief psychiatric rating scale. Psychol. Rep. 10 (3), 799–812.
Paton, C., Whittington, C., Barnes, T.R., 2007. Augmentation with a second antipsychotic in patients with schizophrenia who partially respond to clozapine: a meta-analysis. J. Clin. Psychopharmacol. 27 (2), 198–204.
R Core Team, 2022. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
Repo-Tiihonen, E., Hallikainen, T., Kivisto, P., Tiihonen, J., 2012. Antipsychotic polypharmacy in clozapine resistant schizophrenia: a randomized controlled trial of tapering antipsychotic co-treatment. Ment. Illn. 4 (1), e1.
Samara, M.T., Klupp, E., Helfer, B., Rothe, P.H., Schneider-Thoma, J., Leucht, S., 2018. Increasing antipsychotic dose for non response in schizophrenia. Cochrane Database Syst. Rev. 5 (5). CD011883.
Schmidt-Kraepelin, C., Feyerabend, S., Engelke, C., Riesbeck, M., Meisenzahl-Lechner, E., Verde, P.E., Correll, C.U., Kluge, M., Makiol, C., Neff, A., Lange, C., Englisch, S., Zink, M., Langguth, B., Poeppl, T.B., Reske, D., Gouzoulis-Mayfrank, E., Grunder, G., Hasan, A., Brockhaus-Dumke, A., Jager, M., Baumgartner, J., Leucht, S., Cordes, J., Group, C.S, 2022. Amisulpride and olanzapine combination treatment versus each monotherapy in acutely ill patients with schizophrenia in Germany (COMBINE): a double-blind randomised controlled trial. Lancet Psychiatry 9 (4), 291–306.
Sernyak, M.J., Rosenheck, R., 2004. Clinicians' reasons for antipsychotic coprescribing. J. Clin. Psychiatry 65 (12), 1597–1600.
Shafti, S.S., 2009. Augmentation of olanzapine by fluphenazine decanoate in poorly responsive schizophrenia. Clin. Schizophr. Relat. Psychoses 3 (2), 97–102.
Simpson, G.M., Angus, J.W., 1970. A rating scale for extrapyramidal side effects. Acta Psychiatr Scand. Supplementum 212, 11–19.
Stewart, L.A., Clarke, M., Rovers, M., Riley, R.D., Simmonds, M., Stewart, G., Tierney, J. F., Prisma-IPD Development Group, 2015. Preferred reporting items for systematic review and Meta-analyses of individual participant data: the PRISMA-IPD statement. Jama 313 (16), 1657–1665.
Taylor, D.M., Smith, L., 2009. Augmentation of clozapine with a second antipsychotic–a meta-analysis of randomized, placebo-controlled studies. Acta Psychiatr. Scand. 119 (6), 419–425.
Taylor, D.M., Smith, L., Gee, S.H., Nielsen, J., 2012. Augmentation of clozapine with a second antipsychotic - a meta-analysis. Acta Psychiatr. Scand. 125 (1), 15–24.
Wang, J., Omori, I.M., Fenton, M., Soares, B., 2010. Sulpiride augmentation for schizophrenia. Cochrane Database Syst. Rev. 1, CD008125.
Wang, H., Chen, Y., Lin, Y., Abesig, J., Wu, I.X., Tam, W., 2021. The methodological quality of individual participant data meta-analysis on intervention effects: systematic review. BMJ 373, n736.